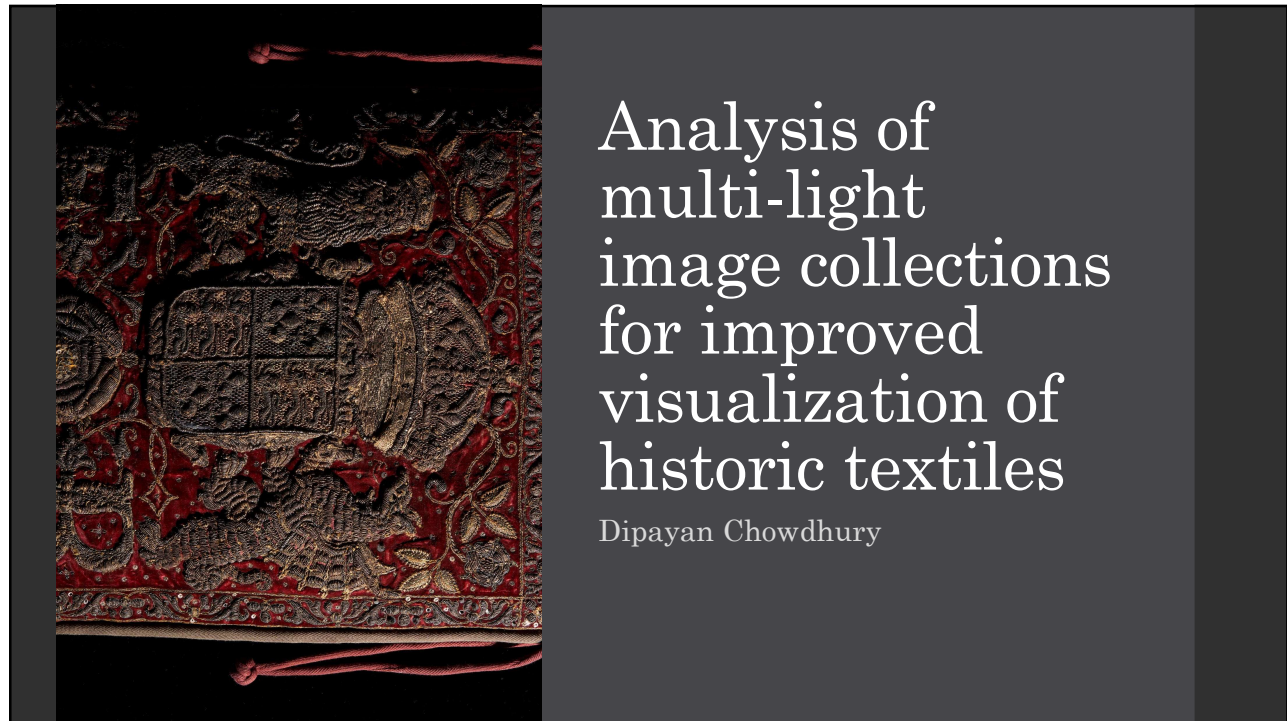




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Summer internship structure

Host organization: NTNU Colourlab

Supervisor: Irina-Mihaela Ciortan

Co-supervisors: Yoko Arteaga; Giorgio Trumpy

Abstract: This study investigates the use of Reflectance Transformation Imaging (RTI) for the analysis of historic textiles, employing the Relight software to generate relightable visualizations from multi-light image collections (MLICs).

Various algorithms within Relight were compared in terms of their ability to accurately reconstruct scene appearance.

Visual inspections and quantitative evaluations using CIELAB color difference formula was performed.

However, challenges such as desaturation in the reconstructed images highlight limitations that require further investigation.

2

Introduction

Multi-light imaging, also known as Reflectance Transformation Imaging (RTI) is a technique for capturing visual data, where the camera and scenes remain static, while the incident light is moving.

It allows for the dynamic relighting of the scene from unrecorded light positions.

Relight is a library to create and view (on the web) relightable images (RTI)

3

Algorithms in Relight

LPTM and PTM –
Polynomial Texture Mapping (18 planes, or 9 using the LPTM where only the luminance change)

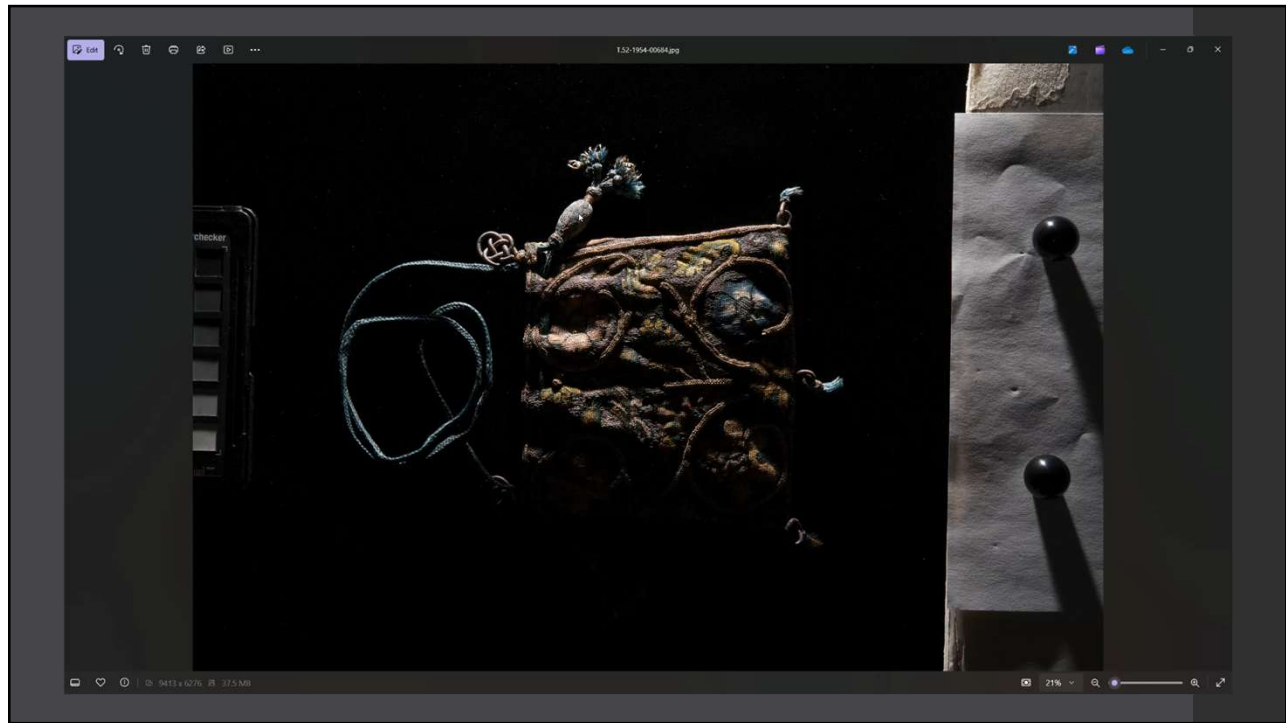
HSB –
Hemispherical Harmonics with 12 or 27 planes

RBF –
Variable number of planes using a sampled basis and radial basis function interpolation.

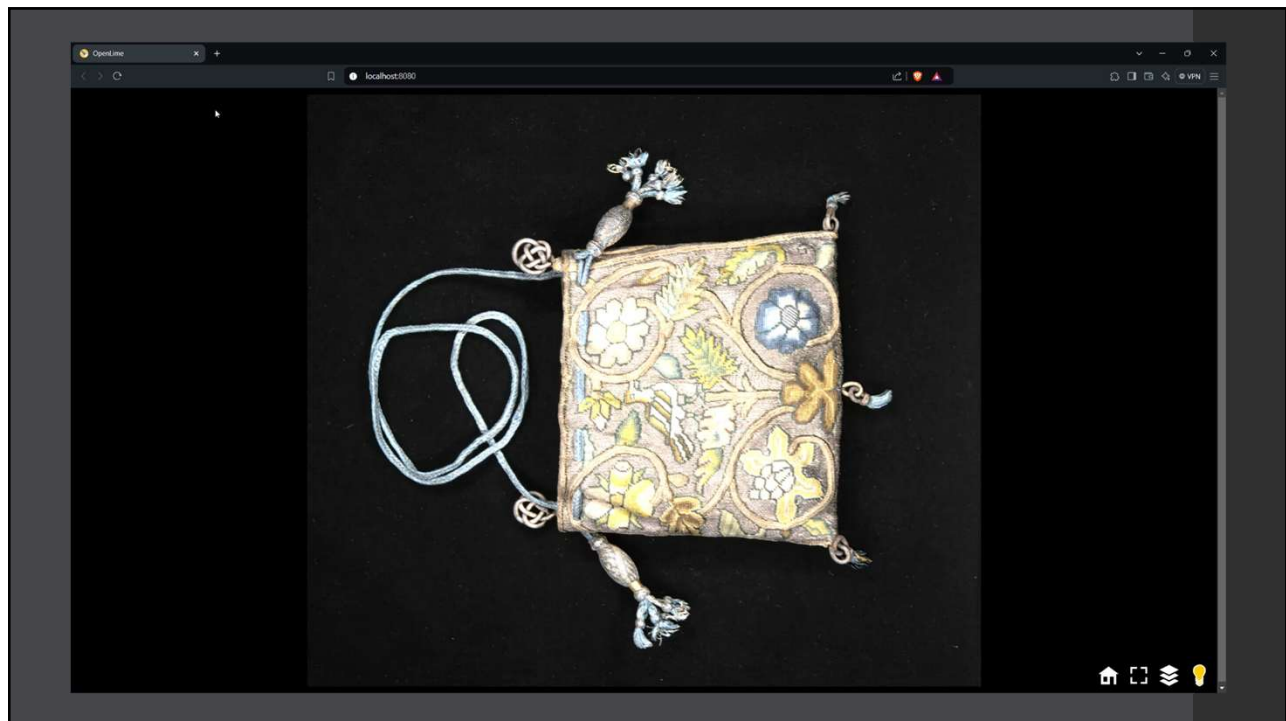
YRBF –
Same as RBF but using the Jpeg color space separately for each channel, resulting in a better chroma fidelity.

BLN and YBLN –
Bilinear interpolation instead of radial basis functions.

4



5



6

Visual Comparison



Original image vs BLN reconstruction

7

Average CIELAB color difference for each image set, calculated by first averaging the pixel-by-pixel color differences between the input and output images for each image, and then averaging those results across the entire image set.

Imageset	Bilinear	LPTM	PTM	RBF	YRBF
T.40 1986 Back	2.8628	3.6128	3.5172	2.486	7.2339
T.40 1986 Front	4.8083	5.054	4.9605	4.2048	6.0964
T.52 1954 Back	5.4108	6.7249	6.5948	4.9935	5.2157
T.52 1954 Front	5.5653	7.1047	6.914	4.9468	5.1407
T.52A 1954 Back	4.5597	6.2149	6.1397	4.0749	4.0478
T.52A 1954 Front	4.3685	5.7805	5.6806	4.0464	4.0199

8

Results and Limitations

Challenge Encountered:

- Significant desaturation of RTIs compared to input images.

Troubleshooting Efforts:

- Explored leave-one-out comparisons.
- Attempted exclusion of shadow-heavy images from the dataset.
- Switched to Linux because the QT commands were not working well in Windows
- Raised multiple issues on Github (still remain unresolved)

Unresolved Issue:

- Persistent desaturation issues – potential software or dataset limitations.

CIELAB Colour Difference Analysis:

- Average CIELAB colour difference calculated for each image set.

9

Future Directions

Main Issue to Address:

- The desaturation problem needs to be explored and resolved, because colour accuracy is important in cultural heritage.
- **Key Questions:** Does the problem lie in the software, dataset, or RTI creation method?

Proposed Actions:

- Conduct rigorous testing of the software using multiple datasets.
- Develop a smooth and efficient workflow.
- Identify potential software improvements (e.g., an option to take snapshots of the RTI
 - within the Relight software).

10

Thank you!

And have a great day :)